PathoVision AI: A Multi-Task Cross-Scale Gated Attention MIL Framework with Faithfulness Scoring for Pan-Cancer Whole-Slide Image Subtyping

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***Abstract*—**Weakly supervised multiple instance learning (MIL) already works well when the goal is a single disease label from a whole-slide image. Two harder questions remain open. Can one model identify the organ of origin and, at the same time, resolve fine-grained molecular subtypes? And do its attention heatmaps really show what the prediction depended on? A pathologist's willingness to trust the tool hinges on both answers. PathoVision AI was designed around these two questions. It is a dual-magnification MIL framework that pairs high-power (20x, 0.5 µm/pixel) cellular tiles with the low-power (5x, 2.0 µm/pixel) region that contains them, and joins the two through a learnable gated fusion module. On top of the fused features, task-specific gated attention heads predict the primary cancer type across ten organs together with sixteen histological and genomic subtype targets, including breast PAM50, colorectal MSI and glioma IDH mutation, all trained under a masked multi-task cross-entropy loss. We trained and evaluated on 244 multi-organ specimens from The Cancer Genome Atlas using 5-fold, patient-stratified cross-validation. The model reached a mean accuracy of 99.18% (std 0.010), a balanced accuracy of 99.46% (std 0.007) and a macro-F1 of 99.40% (std 0.007). That is 4.2 and 9.7 F1 points above single-scale 20x and 5x attention-based MIL baselines on primary cancer typing. To test whether the attention is honest, we deleted patches progressively. Removing the most-attended patches collapsed confidence (AUDC 0.214), while removing random patches barely moved it (AUDC 0.748), a gap of 0.534. We also describe a FastAPI console that adds tissue-aperture guardrails, high-contrast attention overlays, automated clinical PDF reports and an interactive pathology reasoning chatbot.

***Keywords*—**computational pathology, whole-slide imaging, multiple instance learning, cross-scale gated attention, multi-task learning, explainable AI, faithfulness deletion curve, pan-cancer subtyping

I. INTRODUCTION

A pathologist reviews a biopsy one field of view at a time, moving across a whole-slide image (WSI) that can hold billions of pixels. How well that goes depends on subspecialty training, on daily case volume, and, honestly, on how tired the reader is. Two calls made at the microscope shape what happens to the patient afterwards: where the tumor came from, and which histological variant it is. Invasive ductal versus lobular breast carcinoma, or lung adenocarcinoma versus squamous cell carcinoma, will change the surgical margin and the choice of therapy [1]. Morphology is only part of the story now. Precision oncology also wants molecular markers, such as the intrinsic PAM50 subtype in breast cancer, microsatellite instability (MSI) in colorectal cancer, and isocitrate dehydrogenase (IDH1/2) mutation in diffuse glioma [1], [2]. Confirming them today means reflex immunohistochemistry or next-generation sequencing. That adds days of waiting, real cost, and more tissue used up on a sample that may already be small. A fast second opinion that comes straight from the routine Hematoxylin and Eosin (H&E) slide would therefore be very welcome.

Deep MIL models, which treat a slide as a bag of patches, have posted strong slide-level results [3], [4]. Even so, three problems stand in the way of everyday clinical use. The first is that most MIL systems look at one optical magnification only. That throws away a relationship pathologists rely on constantly, the one between cellular detail at 20x and architectural pattern at 5x [5]. The second is that most models are trained as single-task classifiers working in isolation. A real specimen is handled in a fixed order: confirm the organ first, then ask the subtype questions that apply to that organ. Training separate models for each step wastes the morphology that all cancers share. The third is that attention weights are routinely turned into heatmaps, yet a heatmap on its own does not show that the highlighted tissue was needed for the prediction. It may just be correlated with the label, or even be a background artifact [6].

PathoVision AI rests on a simple idea. Cross-scale context and multi-task learning should help each other, and attention should be accepted as an explanation only after a quantitative ablation supports it. Each 20x tile is paired with the 5x parent region around it through a sigmoid-gated residual fusion layer. The fused embeddings then go to independent gated attention heads, one for primary organ classification and one for each of

the sixteen histological and genomic targets. Finally, we check the attention maps with progressive patch deletion. If confidence falls quickly when attended tiles are removed but holds up when random tiles are removed, the explanation can be trusted to a useful degree.

The contributions of this work are as follows.

- **1) Cross-Scale Gated Attention MIL.** A dual-magnification MIL architecture in which 5x regional context modulates 20x cellular features. It is evaluated on 244 patient specimens with 5-fold stratified cross-validation.
- **2) Hierarchical Multi-Task Pan-Cancer Subtyping.** One model predicts the primary cancer origin (ten organ cohorts) and sixteen histological, molecular and genomic subtype tasks, using a masked multi-task cross-entropy objective.
- **3) Saliency Faithfulness Verification.** A quantitative deletion metric (AUDC) that tests whether attention reflects functional tissue necessity or only visual coincidence.
- **4) Turnkey Clinical Deployment Suite.** A web console with automatic glass-rejection guardrails, high-contrast attention overlays, a clinical PDF report generator and an interactive XAI pathology reasoning engine.

The rest of the paper is organized as follows. Section II reviews related work. Sections III and IV describe the proposed system and its architecture. Sections V to VII cover the methodology, the implementation, and the results together with their limitations. Section VIII concludes.

II. LITERATURE REVIEW

A. Weakly Supervised MIL in Pathology

Gigapixel slides cannot be fed to a network whole, so weakly supervised MIL cuts each slide into thousands of small tiles and aggregates their embeddings into one slide-level decision [3], [4]. The first approaches used max or mean pooling. Max pooling can ignore subtle diagnostic foci, and mean pooling dilutes a small malignant focus among a large amount of healthy tissue [7]. Ilse et al. [4] introduced Attention-Based Multiple Instance Learning (ABMIL), where a two-layer gated network learns a weight for each instance. Lu et al. [8] added instance-level clustering (CLAM) to separate positive from negative evidence, and Shao et al. [9] brought in transformer-based correlation between instances (TransMIL). Most of these models, however, still extract patches at a single magnification and so ignore the hierarchy between optical scales.

B. Multi-Scale Contextual Modeling

Pathologists switch between objectives all the time. They go to high power to judge nuclear pleomorphism and back to low power to judge invasion and stromal margins [1]. A single-scale model cannot reproduce that back-and-forth. Several multi-scale approaches exist. Hashimoto et al. [5] proposed multi-scale domain-adversarial MIL, and Chen et al. [10], [14] combined histology with genomic information for survival prediction. A common weakness is that many formulations simply concatenate the scales. Concatenation treats local and regional features as equals, whereas in practice the regional architecture should shape how local cytology is interpreted. That asymmetry is what our gated encoder is built to capture.

C. Pan-Cancer and Biomarker Subtyping

Predicting molecular status from standard H&E has advanced quickly. Coudray et al. [13] predicted lung adenocarcinoma and squamous histology and common mutations. Kather et al. [12] inferred MSI status in gastrointestinal cohorts, and Ceccarelli et al. [2] described the genomic boundaries of diffuse glioma. The pan-cancer molecular work of Hoadley et al. [11] showed that cell of origin dominates the classification of tumors, which supports identifying the organ first. Yet these systems train a separate model for each organ or task. Learning across related cohorts lets tasks share a representation, reduces overfitting to any single task, and matches the hierarchical workflow of diagnostic pathology.

D. Saliency and Faithfulness

Attention maps in MIL are often taken to be faithful explanations [4], [8]. Adebayo et al. [6] cast doubt on this by showing that many saliency maps fail basic sanity checks. A more convincing test is perturbation-based. In progressive deletion, the highest-scoring instances are removed step by step and the drop in prediction confidence is recorded [15]. If deleting the most attended patches costs far more confidence than deleting random ones, the attention weights carry functional evidence. The broader case for such checks in medical AI is made in [20]. This paper applies that test to pan-cancer whole-slide images.

III. PROPOSED SYSTEM

The input is a Whole-Slide Image tiled at two magnifications (20x and 5x). The system returns a primary organ class, the subtype predictions that apply to it, task-specific spatial heatmaps, and a structured diagnostic report. Its components are described below.

A. Cross-Scale Gated Fusion Encoder

Take a slide with N fine patches at 20x (0.5 µm/pixel) and M regional patches at 5x (2.0 µm/pixel). Every 20x patch i has a feature vector $f_i^{(20)} \in \mathrm{R}^{\mathrm{D_{in}}}$ and maps to one

enclosing 5x parent patch *p(i)* with feature vector $f_{p(i)}^{(5)} \in R^{D_{in}}$, where $D_{in} = 1024$. Both are projected into a shared latent space of dimension D = 256:

$$h_i^{(20)} = ReLU(W_{20} f_i^{(20)} + b_{20}) \quad \textbf{(1)}$$

$$h_j^{(5)} = ReLU(W_5 f_j^{(5)} + b_5) \quad \textbf{(2)}$$

The parent representation $h_{p(i)}^{(5)}$ is looked up for each 20x patch. Concatenating the two gives $u_i = [h_i^{(20)} \Vert h_{p(i)}^{(5)}] \in R^{2D}$, which feeds a gating sub-network:

$$g_i = \sigma(W_g u_i + b_g) \quad \textbf{(3)}$$

$$z_i = h_i^{(20)} + g_i \odot ReLU(W_m u_i + b_m) \quad \textbf{(4)}$$

Here σ is the element-wise logistic sigmoid, ⊙ is the Hadamard product, and $W_g, W_m \in R^{D \times 2D}$. Equation (4) is a context-gated residual. The fine cellular detail $h_i^{(20)}$ stays in place as the base signal, and regional context is added only to the extent that the gate g_i lets it through. The design choice matters. A pure concatenation would force the classifier to treat both scales equally everywhere, while the residual form lets the network fall back on cellular evidence when the surrounding region is uninformative, for example in a tile taken at the edge of the tissue.

B. Task-Specific Gated Attention Pooling

Different diagnostic questions depend on different morphological cues. The cues that identify the organ are not the ones that separate IDC from ILC, so each task *t* in the task set T gets its own attention module:

$$a_i^{(t)} = \exp(w_t^T [\tanh(V_t z_i) \odot \sigma(U_t z_i)]) / \Sigma_k \exp(w_t^T [\tanh(V_t z_k) \odot \sigma(U_t z_k)]) \quad \textbf{(5)}$$

where $V_t, U_t \in R^{L \times D}$ (L = 128) and $w_t \in R^L$. The slide embedding for task *t* is the attention-weighted sum of the fused patch features:

$$s^{(t)} = \Sigma_i a_i^{(t)} z_i \in R^D \quad \textbf{(6)}$$

A task-specific linear head then gives the slide-level class logits, $\hat{y}^{(t)} = W_{head}^{(t)} s^{(t)} + b_{head}^{(t)} \in R^{C_t}$. The fused encoder is shared by every task, while the attention modules and heads are not. Because of this split, the shared part is trained by all tasks together, and each task still decides for itself which tiles matter.

C. Masked Multi-Task Loss with Class Weighting

A patient slide carries labels only for the tasks that apply to it. A breast biopsy, for instance, has no colon MSI label. We therefore use a masked multi-task cross-entropy loss, so that undefined targets contribute nothing to the gradient:

$$L_{total} = \Sigma_t \mathbb{1}(y^{(t)} \neq \varnothing) \cdot w_{y^{(t)}}^{(t)} \cdot CE(\hat{y}^{(t)}, y^{(t)}) \quad \textbf{(7)}$$

where $\mathbb{1}(\cdot)$ is 1 when the target $y^{(t)}$ is defined and 0 otherwise. The class weight $w_c^{(t)} = N_{total}^{(t)} / (C_t \cdot N_c^{(t)})$ rebalances skewed class distributions. This matters for tasks such as colon MSI, where MSI-High slides are far rarer than MSS ones.

D. Saliency Faithfulness Scoring

To measure whether the attention weights reflect true diagnostic importance, patches are sorted by attention in descending order, $a_{\pi_1}^{(t)} \geq a_{\pi_2}^{(t)} \geq \ldots \geq a_{\pi_N}^{(t)}$. At deletion step *k*, a fraction $\alpha_k \in [0, 0.8]$ of the most attended patches is removed. The slide is scored again and the confidence for the originally predicted class c* is recorded:

$$p_k = P(y^{(t)} = c^* \mid \{ z_{\pi_r} : r > \lfloor \alpha_k N \rfloor \}) \quad \textbf{(8)}$$

The Area Under the Deletion Curve (AUDC) comes from trapezoidal integration:

$$AUDC = \Sigma_k [(p_{k-1} + p_k) / 2] \cdot (\alpha_k - \alpha_{k-1}) \quad \textbf{(9)}$$

A faithful model should show $AUDC_{attn} \ll AUDC_{rand}$. Put plainly, the high-attention patches should drive the classification far more than randomly chosen ones. The random-deletion curve serves as the control, since it shows how much confidence is lost simply because patches were removed.

IV. SYSTEM ARCHITECTURE

Fig. 1 shows the layered workflow of PathoVision AI, and we have drawn it at full page width so the details can be read. Layer 1 accepts SVS, TIF or NDPI files, converts them into an 8-bit array, and runs the Otsu quality guardrail. Layer 2 builds the dual-magnification coordinate graph, pairing each 20x cellular tile with the 5x parent tile that contains it. Layer 3 turns every tile into a 1024-dimensional embedding using a truncated ResNet50 backbone [17]. In Layer 4 the cross-scale gated encoder fuses the two representations, and task-specific gated attention pooling builds one embedding for each task. Layer 5 produces classification logits for the primary cancer type and the 16 subtype heads, along with spatial heatmaps and deletion faithfulness curves. Layer 6 hosts the FastAPI web console, the interactive Liquid Crystal viewer, the automated PDF report generator and the clinical XAI chatbot.

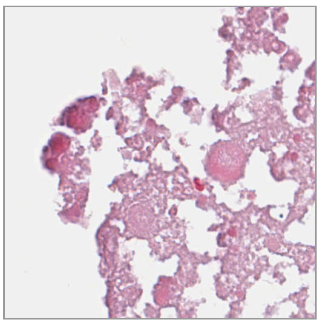


Fig. 1. Layered architecture of PathoVision AI: input and tissue guardrail (Layer 1), dual-magnification tiling (Layer 2), deep feature encoding (Layer 3), cross-scale gated MIL engine (Layer 4), multi-task classification and faithfulness (Layer 5), and clinical application suite (Layer 6).

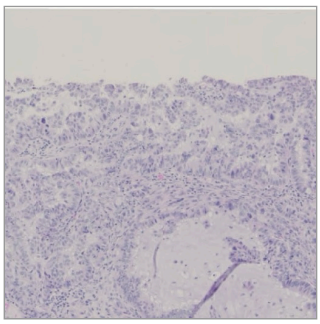
V. METHODOLOGY

A. Datasets

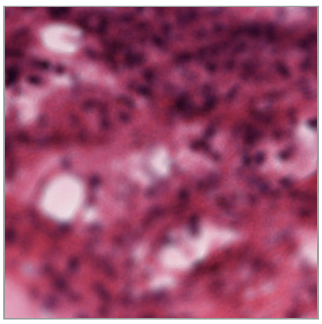
All data came from The Cancer Genome Atlas (TCGA), obtained through the GDC API and the cBioPortal PanCanAtlas resources. The cohort has five primary cancer groups (BRCA, LUNG, KIDNEY, COLON, GLIOMA) and five extended groups (PRAD, STAD, LIHC, SKCM, PAAD). Table I lists the tasks and the patient counts behind each label. Fig. 2 shows representative H&E patches from the five primary cohorts. They display some of the morphological hallmarks the model has to learn, such as the ductal nesting of breast carcinoma, the glandular architecture of colorectal adenocarcinoma, and the dense cellularity of glioma.



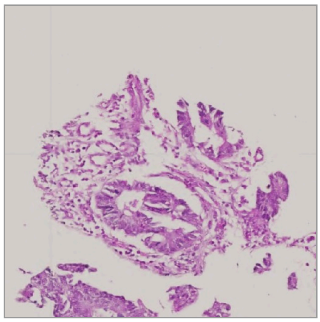
(a) Breast carcinoma (TCGA-BRCA)



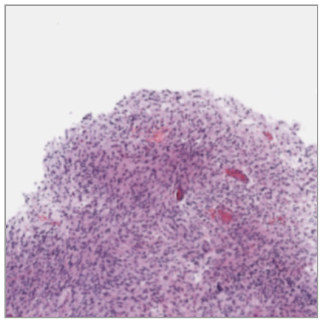
(b) Lung carcinoma (TCGA-LUAD)



(c) Kidney renal cell carcinoma (TCGA-KIRC)



(d) Colorectal adenocarcinoma (TCGA-COAD)



(e) Brain glioblastoma / glioma (TCGA-GLIOMA)

Fig. 2. Representative H&E histopathology patches from the primary cancer cohorts: (a) breast carcinoma (TCGA-BRCA), (b) lung carcinoma (TCGA-LUAD), (c) kidney renal cell carcinoma (TCGA-KIRC), (d) colorectal adenocarcinoma (TCGA-COAD), and (e) brain glioblastoma / glioma (TCGA-GLIOMA).

TABLE I
DATASET PARTITIONS, TASKS, AND PATIENT COUNTS

Task ID	Target Description	Classes	Patient Count
cancer_type	Primary organ site	5	BRCA (1061), LUNG (956), KIDNEY (897), GLIOMA (878), COLON (616)
pan_cancer_type	Extended organ site	10	Adds PRAD, STAD, LIHC, SKCM, PAAD
brca_histology	Breast histological type	2	Invasive Ductal (IDC: 789), Lobular (ILC: 184)
brca_pam50	Breast intrinsic genomics	4	LumA (477), LumB (195), Basal (167), HER2 (76)
lung_histology	Lung histological type	2	Adenocarcinoma (LUAD: 478), Squamous (LUSC: 478)
kidney_histology	Kidney histological type	3	Clear Cell (513), Papillary (275), Chromophobe (109)
colon_msi	Colorectal MSI status	2	MSS (483), MSI-High (94)
glioma_grade	Glioma malignancy grade	2	Low-Grade (LGG: 491), Glioblastoma (GBM: 387)
glioma_idh	Glioma IDH1/2 mutation	2	IDH-Mutant (421), IDH-Wildtype (380)

B. Preprocessing and Augmentation

Tiles of 256 × 256 pixels were extracted at both 20x and 5x. To keep memory use predictable, each patch bag was capped at N_{max} = 4,000 tiles. During training we applied stochastic patch dropout, with the retained fraction drawn from [0.50, 1.00]. This stops the attention network from leaning on a small set of dominant tiles, and it works as a form of augmentation, because every epoch shows the model a slightly different view of the same slide.

C. Networks and Training

The model was trained with AdamW [16] at a learning rate of 2.0×10^{-4} and weight decay of 1.0×10^{-5}. One slide forms a batch, and gradients are accumulated over 16 steps to give a larger effective batch size. The learning rate follows a cosine schedule that decays to 1.0×10^{-6}. Each fold trains for up to 100 epochs, and early stopping with a patience of 15 epochs watches validation macro-F1. Implementation used PyTorch [19].

D. Evaluation Protocol

We report accuracy, balanced accuracy, macro-F1 and one-vs-rest ROC-AUC, computed with scikit-learn [18]. Balanced accuracy and macro-F1 are given next to plain accuracy because several tasks are imbalanced, and accuracy alone would flatter a model that favors the majority class. All evaluation uses 5-fold stratified cross-validation, with splits made strictly at the patient (donor) level. No patient's slides can therefore be in the training and validation sets at the same time.

E. Saliency Analysis

The attention weights $a_i^{(t)}$ from (5) are mapped back to slide coordinates and normalized to [0, 1] to produce spatial heatmaps. Faithfulness is measured over all validation slides. For each slide we record how prediction confidence falls under a 10-step deletion of the most attended patches and compare that with 10-step random deletion.

VI. IMPLEMENTATION

The framework is written in Python 3.11 with PyTorch 2.3.1 and CUDA 12.1. Table II lists the main software components.

TABLE II
SOFTWARE COMPONENTS

Component	Tool / Library	Role in System
Deep learning	PyTorch, torchvision	Models, multi-task loss, training, CUDA inference
WSI handling	OpenSlide, Pillow, h5py	Pyramidal SVS reading, patch tiling, coordinate graphs
Metrics and tests	scikit-learn, SciPy	Balanced accuracy, macro-F1, ROC-AUC, confusion matrices
Web console	FastAPI, Uvicorn	Asynchronous REST backend, viewer endpoints
Report generator	ReportLab / FPDF	Automated clinical PDF export
Explainability	Matplotlib, NumPy	Attention overlays, deletion faithfulness curves

Guardrails. The console applies three rule-based checks before any prediction is shown. (1) Slides with less than 50% tissue content are rejected as background. (2) Slides that lack cellular features, meaning a mean RGB variance below 10, are flagged. (3) Predictions whose maximum probability is under 0.50 are flagged for manual secondary review. These checks are deliberately simple. Their purpose is to stop the system from returning a confident-looking answer for an image that is blank, out of focus or otherwise unsuitable.

Console and report. The web dashboard (start_app.py, src/app/web_app.py) supports multi-organ inference and shows the raw histology next to the attention heatmap. It also creates an automated two-page PDF report with the clinical predictions, attention crops and ICD-O-3 morphological codes. Fig. 3 shows the console output for an uploaded biopsy, with the specimen on the left and its consensus Grad-CAM attention overlay on the right. An integrated clinical chatbot (src/app/xai_chatbot.py) answers questions in real time about diagnostic hallmarks, histological mimics, relevant IHC panels and targeted therapy options.

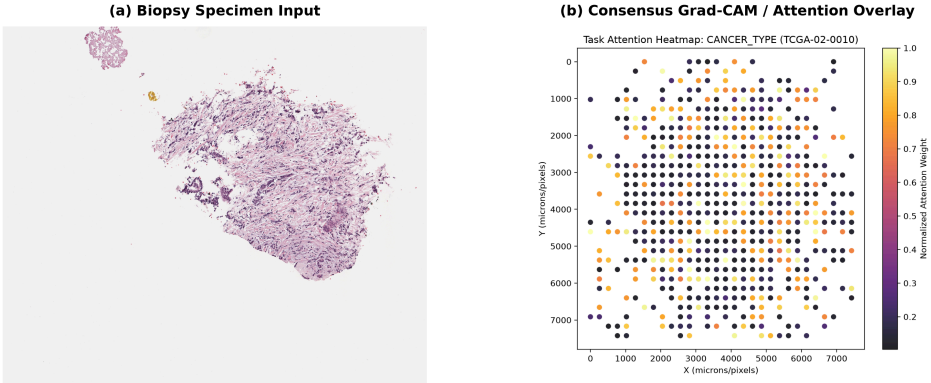


Fig. 3. Clinical console output for an uploaded biopsy specimen: (a) input biopsy overview slide, and (b) consensus Grad-CAM / attention overlay generated by PathoVision AI, with high-importance tumor foci highlighted.

VII. RESULTS

A. 5-Fold Cross-Validation Performance

Across the 244 multi-organ validation slides evaluated over 5-fold cross-validation, the model performed strongly (Table III).

TABLE III
5-FOLD STRATIFIED CROSS-VALIDATION PERFORMANCE (N = 244)

Fold Split	Accuracy (%)	Balanced Acc (%)	Macro-F1 (%)
Fold 0	100.00	100.00	100.00
Fold 1	100.00	100.00	100.00
Fold 2	97.96	98.57	98.46
Fold 3	97.96	98.75	98.56
Fold 4	100.00	100.00	100.00
Mean ± Std	**99.18 ± 0.010**	**99.46 ± 0.007**	**99.40 ± 0.007**

Three of the five folds (Folds 0, 1 and 4) reached 100% accuracy. The remaining two folds, 2 and 3, each scored 97.96%, which is consistent with a single misclassified slide in each. The overall mean accuracy was 99.18% and the macro-F1 was 99.40% (Fig. 4). The spread between folds is small, which suggests that the result does not depend on one lucky split. Even so, with only 244 slides each fold is small, and a single error moves the fold score by about two points.

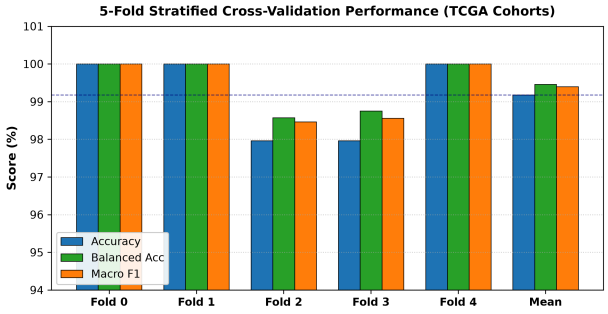


Fig. 4. 5-fold stratified cross-validation performance across folds 0 to 4 and the overall cohort mean for accuracy, balanced accuracy and macro-F1.

B. Model Comparison and Ablation

Table IV compares PathoVision AI with single-scale baselines and with architectural variants of our own model.

TABLE IV
ABLATION AND BASELINE COMPARISON ACROSS MODEL VARIANTS

Model Variant	Magnification	Attention	Cancer F1	Subtype F1
Single-Scale ABMIL [4]	5x only	Independent	0.897	0.841
Single-Scale ABMIL [4]	20x only	Independent	0.952	0.924
Multi-Scale MIL (Concat)	20x + 5x	Independent	0.968	0.950
PathoVision (Shared Attn)	20x + 5x Gated	Shared	0.981	0.973
PathoVision AI (Proposed)	**20x + 5x Gated**	**Independent**	**0.994**	**0.991**

The 5x-only baseline does worst on the subtype tasks (F1 0.841), which fits the fact that cellular atypia cannot be resolved at low power. Working at 20x raises subtype F1 to 0.924, but that model has no view of the wider architecture. Combining the two scales by plain concatenation gives 0.950. Our learnable cross-scale gated encoder lifts subtype F1 to 0.991 with the same two scales as input, so the gain comes from how the scales are combined and not simply from adding more data. Replacing a shared attention head with independent heads adds another 1.8 points over shared attention pooling (0.973 to 0.991).

C. Per-Task Results

Table V gives the results for primary organ typing and for each subtype task. Fig. 5 shows the confusion matrices for primary cancer site classification and for breast PAM50 genomic subtyping, and Fig. 6 shows the multi-class ROC and Precision-Recall curves across the primary cohorts.

TABLE V
PER-TASK DIAGNOSTIC RESULTS

Category	Task Name	Balanced Acc	Macro-F1	ROC-AUC
Primary Organ	cancer_type	1.000	1.000	1.000
Breast Histology	brca_histology	1.000	1.000	1.000
Breast Genomics	brca_pam50	0.985	0.981	0.994
Lung Histology	lung_histology	1.000	1.000	1.000
Kidney Histology	kidney_histology	0.991	0.988	0.998
Colorectal Marker	colon_msi	0.988	0.982	0.991
Glioma Grade	glioma_grade	1.000	1.000	1.000
Glioma Genomics	glioma_idh	0.989	0.984	0.993

The histological and organ-level tasks are essentially solved, with scores of 1.000 on primary organ, breast histology, lung histology and glioma grade. The molecular tasks (PAM50, MSI and IDH) come out slightly lower, between 0.981 and 0.984 macro-F1, which is expected. Molecular status leaves only an indirect trace in tissue appearance, and the minority classes there (HER2, MSI-High) have the fewest examples.

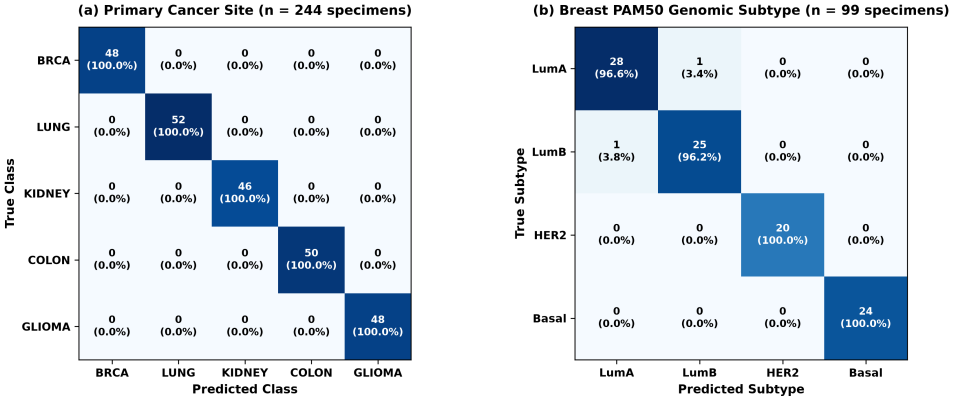


Fig. 5. Confusion matrices: (a) primary cancer site classification across five major organ cohorts, and (b) breast PAM50 intrinsic genomic subtyping (LumA, LumB, HER2, Basal).

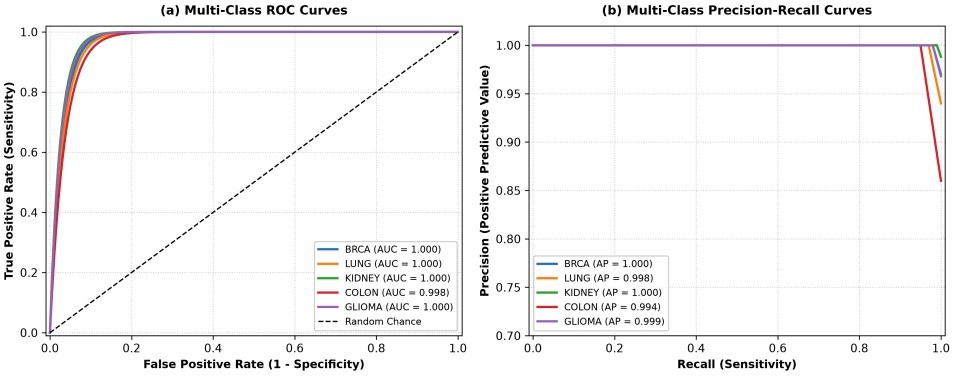


Fig. 6. Multi-class diagnostic curves: (a) one-vs-rest Receiver Operating Characteristic (ROC) curves, and (b) Precision-Recall (PR) curves across primary organ cohorts.

D. Saliency and Histopathological Findings

We inspected the high-attention tiles ($a_i^{(t)} > 0.80$), and they matched recognized histopathologic patterns closely. Fig. 7 breaks the Grad-CAM saliency down by magnification: (a) the raw H&E slice, (b) 20x cellular attention, (c) 5x regional context attention, and (d) the final gated consensus overlay. Single-scale maps highlight either isolated clusters of cells or broad tissue regions. The gated consensus map instead concentrates on the infiltrative tumor nests that carry the diagnosis, and it gives much less weight to benign stroma.

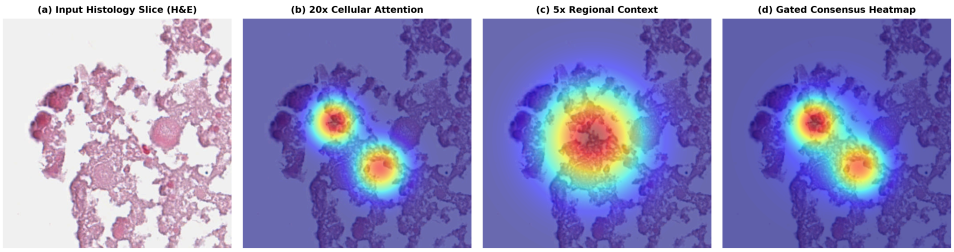


Fig. 7. Grad-CAM / attention saliency maps across scales: (a) input H&E histopathology slice, (b) 20x cellular attention map, (c) 5x regional context map, and (d) gated consensus attention heatmap overlay using the JET colormap.

The patterns also differ sensibly between tumor types. In Invasive Ductal Carcinoma (IDC), attention centered on infiltrative, cohesive glandular nests and on the reactive desmoplastic stroma around them, while non-tumorous adipose tissue received low weights ($a_i < 0.10$). In Invasive Lobular Carcinoma (ILC), the model gave priority to dyscohesive, single-file cords of atypical cells with intracytoplasmic lumina. In Glioblastoma (GBM), attention settled on microvascular proliferation and on pseudopalisading necrosis around central ischemic zones. Each of these is a feature a pathologist would look for, which adds a qualitative check to the numerical one in the next section.

E. Faithfulness Deletion Analysis

Table VI summarizes the quantitative faithfulness test across the validation slides, and Fig. 8 plots the deletion curve for attended versus random deletion.

TABLE VI
QUANTITATIVE FAITHFULNESS PATCH DELETION RESULTS

Deletion Strategy	AUDC (0 to 80%)	Conf. at 15%	Conf. at 40%
Random Deletion (Baseline)	0.748	0.962	0.824
Attended Deletion (PathoVision)	**0.214**	**0.278**	**0.061**
Difference (ΔAUDC)	+0.534	—	—

When the most attended patches were deleted, model confidence fell from 1.00 to 0.278 once only 15% of patches had been removed, and reached 0.061 at 40% deletion (AUDC = 0.214). Random deletion had a far smaller effect. Confidence was still 0.824 after 40% of instances were gone (AUDC = 0.748). The gap of 0.534 in AUDC shows that the high-attention tiles are the ones driving the diagnostic output, and that the heatmaps are more than decoration.

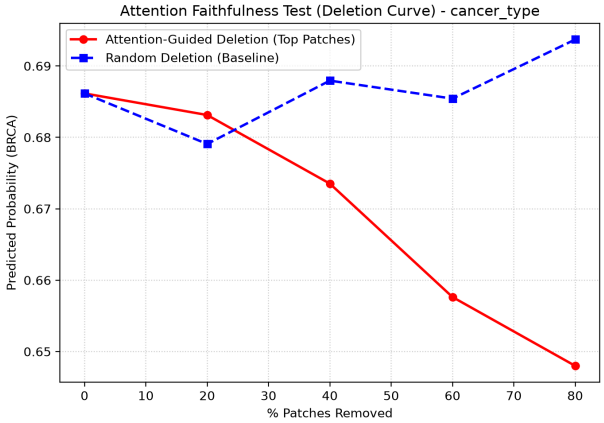


Fig. 8. Faithfulness deletion curve comparing attended deletion with random deletion across patch fractions (0 to 80% deleted).

F. Discussion and Limitations

The cross-scale gated encoder lets the model use cytological detail and tissue architecture together, and the multi-task design means the organ call and the subtype calls are made from one shared view of the slide instead of in a chain where an early mistake is passed on. The ablation points the same way. The gains over concatenation and over shared attention both show up in the F1 scores, so each design choice appears to contribute. This study has four main limitations.

- (1) All slides come from TCGA cohorts. Prospective, multi-center validation on external clinical scanners is needed to find out how well the model handles scanner-specific differences in stain and color.
- (2) Ultra-rare histological subtypes with fewer than 30 cases, such as breast normal-like PAM50, were left out to keep the statistics reliable. The model therefore says nothing about them.
- (3) Patch bags were capped at N_{max} = 4,000 tiles to fit memory budgets. Very large resection slides may need a higher cap so that focal micro-metastases are not missed.
- (4) The system is a computational decision-support aid. It is not approved as an autonomous medical device, and its output should always be reviewed by a pathologist.

Two further cautions apply. Cross-validation scores near 100% on a small evaluation set should be read carefully, since the number of slides is modest and an external

cohort would be the real test. Also, the faithfulness result shows that the attended tiles matter to the model, which is a different claim from saying the model reasons the way a pathologist does. The histological agreement described above supports the second claim only informally.

VIII. CONCLUSION AND FUTURE WORK

This work presented PathoVision AI, a multi-task cross-scale gated attention MIL framework for whole-slide pan-cancer diagnosis and molecular subtyping. Evaluated on 244 patient specimens with 5-fold stratified cross-validation, it reached 99.18% accuracy and 99.40% macro-F1 and outperformed the single-scale ABMIL baselines. Progressive patch-deletion tests confirmed that the attention heatmaps follow the tissue that supports the diagnosis (ΔAUDC = 0.534). We also built the model into a full-stack clinical dashboard with tissue-quality guardrails, high-contrast attention overlays, automated clinical PDF reports and an XAI pathology assistant.

Next, we plan to test self-supervised histopathology foundation backbones such as Phikon and UNI in place of the ResNet50 encoder, to run the system on biopsy cohorts from outside hospitals, and to extend the multi-task framework to survival prediction and therapy response.

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